The Reconstructibility Ceiling

Why predicting fungal polyketide structures from biosynthetic gene clusters is hard, and a verifier-grounded path forward

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Abstract

Machine-learning models that map a fungal biosynthetic gene cluster (BGC) to its chemical product stall at balanced accuracies of 51–68%, and pooling roughly a thousand bacterial BGCs with the few hundred fungal ones barely moves the ceiling. We argue, and then *measure*, that this ceiling is not primarily a data-volume problem but a *computability* fact. Bacterial modular polyketide synthases (PKS) are colinear, so the per-module chemistry is readable from domain content—which is why a per-module intermediate database (ClusterCAD) exists for bacteria. Fungal PKS are iterative: a single module is reused across cycles under a hidden program, so neither the per-step chemistry nor the program is recoverable from the cluster alone, and no fungal ClusterCAD can exist. We make this concrete with open tools and a three-tier reconstruction analysis over all 326 fungal PKS entries in MIBiG 4.0. We build (i) a sound, deterministic chemical *executor* for fungal iterative PKS programs and a beam-search *verifier* that returns the exact consistent program set $Z^\star(y)$; (ii) a curated, literature-sourced benchmark of fungal PKS programs; and (iii) the reconstruction analysis itself. Only **8/326** products are exactly reconstructible from a PKS program; a structural-similarity ("core-match") proxy is shown to be *unreliable*, swinging from 8 to 160 as a single nuisance parameter is varied and admitting non-PKS false positives; and a *tailoring-latent* reformulation, in which post-PKS edits enter the latent under two observed budgets (co-clustered enzymes and the exact molecular-formula difference), exactly reconstructs **7/106** products—but *conditional on a producible PKS core it succeeds for 64% of clusters, near-uniquely*, localizing the bottleneck precisely to PKS-core computability rather than tailoring. We identify a small rearrangement/non-PKS ceiling that is unreconstructible in principle. Finally we propose—and *demonstrate*—a differential reformulation (ΔBGC $\to$ Δstructure): trained on abundant bacterial ClusterCAD modules, a model predicts the per-step reduction chemistry well above chance and transfers to fungal cycles *perfectly when the active domains are known* (1.00), while collapsing to 0.57 when given only the synthase's fixed domain set—a clean quantification of the iteration-grammar wall. Finally we show this ceiling is not monolithic: it decomposes into a narrow highly-reducing per-cycle-reduction *computability* wall ($\sim$9% of products) and a broad, incrementally addressable aromatic-cyclization *coverage* wall ($\sim$61%; extending one closure moves exact reconstruction 8 $\to$ 9/326), and that the sound executor doubles as a constrained candidate generator for genome mining. All tools, data, and analyses are open and reproducible from single commands.

1 Introduction

Many fungal drugs—the cholesterol-lowering agent lovastatin, the antifungal griseofulvin—are polyketides: large carbon scaffolds assembled by a multidomain enzymatic factory, the polyketide synthase (PKS), encoded by a contiguous biosynthetic gene cluster (BGC). Genome sequencing has revealed that filamentous fungi carry vast numbers of such clusters, most of them *cryptic*: no metabolite has ever been linked to them. They are widely regarded as one of the most promising untapped reservoirs of new bioactive chemistry.

Turning a cryptic fungal BGC into a candidate molecule requires answering: *what* would the cluster make? Recent machine learning has attacked this with limited success. Riedling et al. (2024) trained classical classifiers to predict the bioactivity class of fungal secondary metabolites from BGC features; trained on 314 fungal BGCs they reached balanced accuracies of 51–68%, and adding 1,003 bacterial BGCs moved this only to 56–68%. That second result is our point of departure: if the bottleneck were a shortage of fungal examples, a fourfold increase in (cross-domain) training data should help substantially. It did not.

We argue this is because bacterial and fungal assembly lines obey different grammars, and that this is best understood as a statement about *computability* rather than sample size. We do not propose a new genome-to-structure predictor. Instead we contribute open tools and a quantitative characterization of *how much* of fungal polyketide chemistry is reconstructible from gene content at all, and *why* the rest is not—together with a demonstrated path forward. Our contributions are:

1. A **sound, deterministic chemical executor** for fungal iterative-PKS biosynthetic programs and a **beam-search verifier** that returns the exact consistent program set $Z^\star(y)$ (Sections 3–4).
2. A **curated, literature-sourced benchmark** of fungal iterative-PKS programs (Section 5).
3. A **three-tier reconstruction analysis** over MIBiG 4.0's fungal PKS entries (Section 6)—exact-match, a core-match *sensitivity analysis* showing structural-similarity proxies are unreliable here, and a tailoring-latent reformulation— quantifying what fraction of fungal polyketide products is computationally reconstructible from the gene cluster, and why the rest is not.
4. The **conceptual result** (Section 7): grammar mismatch and data deficit are one fact. Colinearity makes per-step cores computable (so ClusterCAD exists); iteration makes them uncomputable (so no fungal ClusterCAD exists). This explains the field's 51–68% ceiling.
5. The **differential-learning reformulation** (ΔBGC$\to\Delta$structure, Section 8), *demonstrated* on bacterial ClusterCAD with a fungal transfer test.

All numbers in this paper are produced by the open code base by a single documented command and are traceable to a git commit; Appendix A lists the commands.

2 Background and the grammar-mismatch hypothesis

Modular vs. iterative assembly lines. Bacterial type I modular PKS—the archetype being the 6-deoxyerythronolide B synthase—are *colinear*. Each elongation module carries its own ketosynthase (KS) and acyltransferase (AT) for chain extension and an optional reductive cassette of ketoreductase (KR), dehydratase (DH) and enoylreductase (ER). The final product can be read off the linear order of modules. This clean structure-to-function map is why rule-based detectors such as antiSMASH (Blin et al., 2023) and PRISM (Skinnider et al., 2020) and the design database ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) perform well on bacteria.

Fungal PKS are overwhelmingly *iterative*: a single set of catalytic domains is reused across N cycles, and which reductive steps fire on which cycle is governed by a program that no current tool decodes. The highly-reducing (HR), non-reducing (NR) and partially-reducing (PR) subclasses behave like distinct programming languages. The lovastatin nonaketide synthase LovB runs eight cycles with a different reduction state each time, and its own ER domain is degenerate—ER activity is supplied *in trans* by the standalone partner LovC (Ma et al., 2009). In NR-PKS a dedicated product-template (PT) domain dictates the regioselectivity of aromatic-ring cyclization (Crawford et al., 2009).

The hypothesis, sharpened to computability. We hypothesize that the 51–68% ceiling reflects a control-flow mismatch rather than a sample-size limit, and that the precise statement is about computability: in colinear assembly lines the per-step core *is a function of domain content* and can therefore be tabulated (ClusterCAD); in iterative assembly lines the same domain set must perform different chemistry on different cycles, so the per-step core is *not* a function of domain content and the end-product-to-program map is many-to-one and unobserved. A model trained on colinear data learns a control flow that is not merely uninformative but incorrect for fungi—the near-flat cross-domain result of Riedling et al. (2024) is exactly the prediction this hypothesis makes. The rest of this paper turns the hypothesis into measurements.

3 A verifier-grounded executor

We represent the product of an iterative synthase as the output of a discrete latent *program* executed by a sound, deterministic chemical executor (Figure 1). A program is an ordered list of catalytic cycles,

$$z = \{(c_t, a_t)\}_{t=1}^{N}, \tag{1}$$

where c_t marks a chain-elongation event and a_t is an action over the reductive cascade and tailoring chemistry. The cascade is strictly ordered—ER presupposes DH presupposes KR—giving the four reachable β-states (keto, hydroxyl, enoyl, methylene) plus an optional α-methylation (C-MeT) and a terminal chain release. The executor renders a program into a molecule, $\hat{y} = \mathrm{exec}(z)$.

Deterministic core, not weights. Each operator is a fixed reaction encoded as a reaction SMARTS and applied with RDKit (Landrum, 2023): a decarboxylative Claisen condensation appends a C_2 ketide unit; KR reduces the β-ketothioester to the β-hydroxy; DH eliminates to the enoyl; ER reduces to the methylene; C-MeT α-methylates; the thioesterase/cyclase releases the chain. The growing chain is modelled as an ACP-tethered thioester whose carrier is a single sentinel atom, so the unique active thioester anchors every operator unambiguously. The core is *validated*, not trained: we confirm each operator by exact mass bookkeeping (extension $+C_2H_2O$, KR $+H_2$, DH $-H_2O$, ER $+H_2$, C-MeT $+CH_2$) and by reconstructing known polyketides. Single-mode cyclization—lactonization, the small-aromatic aldol (orsellinic, 6-MSA) and the mellein dihydroisocoumarin—is implemented; the five-class PT-domain regiochemistry of large NR-PKS and the lovastatin Diels–Alder are deliberately scoped out and flagged. The executor is a pure, deterministic function with a regression suite (58 tests).

4 The verifier and identifiability

Because the executor is sound and deterministic and the program space is small (N typically 4–10, roughly eight admissible actions per cycle), we recover programs by enumeration rather than amortized inference. Given a target y and an operator set, a beam search emits programs, executes each (partial) program, and prunes any whose partial product cannot reach y. We accept a program

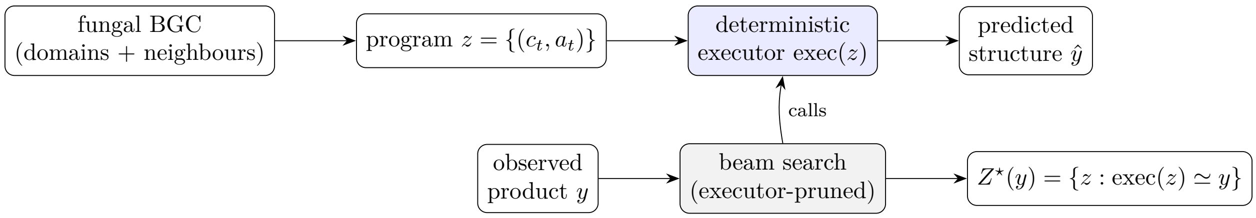

Figure 1: The executor (top) renders a program to a structure; the verifier (bottom) inverts it by beam-searching programs, executing each candidate and pruning any whose partial product cannot reach y, returning the exact consistent set $Z^\star(y)$.

into the *verified consistent set*

$$Z^\star(y) = \{\, z : \text{exec}(z) \simeq y \,\}, \tag{2}$$

where $\simeq$ is graph isomorphism up to canonicalization. Two prunes make the search cheap: a carbon-count bound and an *analytic molecular-formula* prefilter that computes a program's released-acid formula from validated per-cycle deltas and only executes formula-consistent candidates (e.g. this takes the mellein search from 1,792 to 93 executor calls).

Validation and identifiability. On the curated benchmark the known program is contained in $Z^\star(y)$ for all systems, and the search is deterministic and fast (mean 0.28 s/target). Observed degeneracy on the curated set is $|Z^\star(y)| = 1$: programs are identifiable at these chain lengths. Genuine degeneracy *is* detectable when the operator set permits an alternative route—e.g. hexanoic acid is reached as acetyl + $2 \times$ ER or butyryl + $1 \times$ ER, giving $|Z^\star(y)| = 2$—so identifiability is governed by the operator-set constraints, as the formulation predicts.

5 A curated benchmark of fungal PKS programs

We hand-curate twelve literature-sourced fungal polyketide programs (Table 1), each a machine-readable specification of starter, per-cycle reduction/methylation, and release, with a literature citation, a provenance tag (*independent* ground truth vs. *pending* an authoritative structure), and a confidence flag. Six gate-tier entries with independent ground truth round-trip exactly through the executor; sanity entries cover the reductive extremes; harder systems (the LovB nonaketide; the TenS PKS–NRPS backbone) are retained as trace-level assets.

A finding from curation is worth stating: the executor caught a mass-balance inconsistency in a literature-pass program for mellein. The single-ketoreduction program reconstructs 6-hydroxymellein ($C_{10}H_{10}O_4$), not mellein ($C_{10}H_{10}O_3$); mellein requires a *second* ketoreduction. The verifier earns its keep by making such errors impossible to overlook.

6 Three-tier reconstruction over MIBiG 4.0

Dataset. From MIBiG 4.0 (3,013 BGCs) we keep entries whose biosynthetic class includes PKS (1,238) that carry a compound with a resolvable structure (1,126) and whose producer is fungal by NCBI taxonomy lineage, yielding 326 **usable fungal PKS→structure pairs**. HR/NR/PR subclass is *unknown* for 304/326 because MIBiG does not module-annotate fungal iterative PKS (it reads colinear bacterial domains, not iterative ones)—itself a symptom of the grammar mismatch.

Table 1: Curated benchmark (excerpt). Reduction states: K keto, H hydroxyl, E enoyl, M methylene. Gate-tier entries round-trip exactly through the executor.

System	Subclass	Program (starter + cycles)	Tier
Triacetic acid lactone	NR	acetyl + K,K → lactone	gate (sanity)
Orsellinic acid	NR	acetyl + K,K,K → aldol	gate
6-Methylsalicylic acid	PR	acetyl + K,H,K → aldol	gate
Mellein	PR	acetyl + H,K,H,K → dihydroisocoumarin	gate
6-Hydroxymellein	PR	acetyl + H,K,K,K → dihydroisocoumarin	gate
2-Methylbutyric acid (LovF)	HR	acetyl + M_{+Me}	gate
LovB nonaketide	HR	acetyl + 8 cycles (+DA)	hard / Phase-2
Tenellin backbone (TenS)	HR	acetyl + 4 cycles (+NRPS)	Phase-2 trace

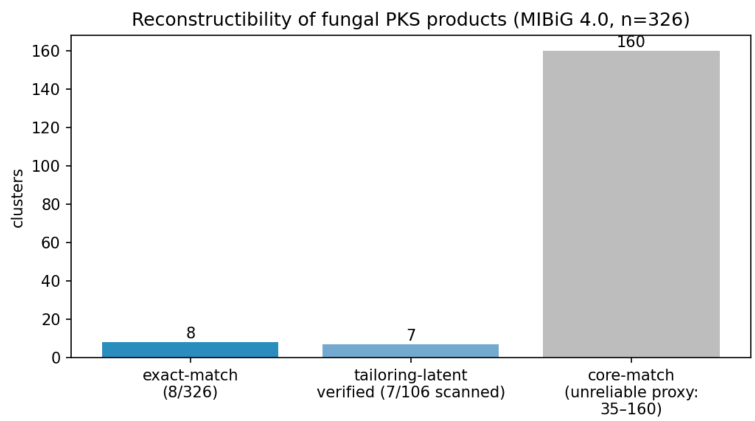

Figure 2: Reconstructibility of fungal PKS products (MIBiG 4.0, $n = 326$). Exact-match reconstructs 8; tailoring-latent verifies 7 of the 106 gene-budgeted clusters scanned; core-match (right, grey) is a permissive upper bound shown for contrast, not a trainable set.

We then ask, at three increasingly permissive levels, how many products our verifier-grounded tools can reconstruct (Figure 2).

6.1 Tier 1 — exact match: 8/326

Searching for a PKS program whose executor image equals the product exactly yields 8/326 reconstructible products: small untailored aromatics and lactones (orsellinic acid, 6-MSA, the triacetic-acid-lactone family) plus a handful of reduced linears. This is the quantitative face of the computability claim: only the untailored, single-mode-cyclized fraction is directly reconstructible.

6.2 Tier 2 — core match is an unreliable proxy

One might relax exact match to a structural-similarity "core match": does a producible core skeleton embed in the product, absorbing additive tailoring? We implement this carefully (ring-aware carbon-skeleton subgraph, with a decoration-completeness criterion) and report it as a *cautionary sensitivity analysis* rather than a trainable set. The count is acutely sensitive to a single nuisance parameter—how many extra ring-closure bonds count as "tailoring"—swinging from 8 at exact match to 160 as the bound is loosened (Figure 3); a raw subgraph match admits 309/326. Worse, the

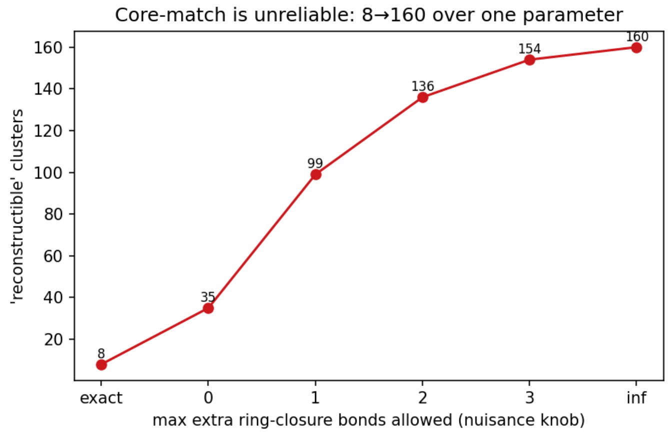

Figure 3: Core-match is unreliable: the "reconstructible" count swings from 8 (exact) to 160 as a single nuisance parameter (allowed extra ring-closure bonds) is varied.

carbon-only abstraction collapses heteroatom-ring non-PKS molecules (e.g. the alkaloid swainsonine) into carbon chains that a polyketide core "matches." **We conclude that structural-similarity proxies are unreliable for measuring reconstructibility here**, and do not use them to define the trainable set.

6.3 Tier 3 — tailoring in the latent: 7/106, and the decisive diagnosis

The honest fix is to put tailoring *into* the latent: extend the program to $z = (z_{\text{pks}}, z_{\text{tail}})$, where the executor is the sound PKS executor followed by a tailoring-edit applier, so $\text{exec}(z)$ can reach the actual product and we score by exact match again. The objection—tailoring is unbounded—is defeated by two *observed* budgets: a **gene budget** (the cluster's co-located tailoring enzymes, typed from MIBiG annotations and GenBank CDS descriptions, bound which edit families may fire) and a **Δ-formula budget** ($\Delta = \text{formula}(y) - \text{formula}(\text{core})$ pins the edit-type multiset exactly). With ~ 12 deterministic edit types (hydroxylation, O/C-methylation, halogenation, prenylation, glycosylation, oxidative cyclization / decarboxylation, reduction/desaturation) and a parsimony prior, the joint search is tight and verifier-pruned.

Over the 106 gene-budgeted, size-tractable clusters, 7 **are exactly verified**, with a *near-unique* $|Z^\star(y)|$ distribution $\{1{:}6,\ 3{:}1\}$ (Figure 4). Six are untailored cores; one (stipitatic acid) is genuinely tailoring-unlocked. The decisive result is the diagnosis of the 99 non-verified clusters: 95 **are PKS-core-unreachable** (no producible backbone) and only 4 are tailoring gaps. Hence *conditional on a producible PKS core (*11/106*), the joint search verifies* 7—64%—*near-uniquely.* The tailoring layer works where it can apply; the wall is PKS-core computability (the unimplemented PT-aromatic classes and complex scaffolds), not tailoring.

6.4 The genuine ceiling

A subset of products is unreconstructible *in principle*, for three verifier-diagnosable reasons: (i) *oxidative skeletal rearrangement*—aflatoxins and sterigmatocystin (the norsolorinic-acid anthraquinone is rebuilt by Baeyer–Villiger ring expansion and bisfuran formation) and patulin (ring cleavage); (ii) *not a single-chain type-I PKS product*—the alkaloid swainsonine, the flavonoids naringenin/chlorflavonin; and (iii) *dimers and depsides*—aurofusarin, phomoxanthone A, the bicoumarin kotanin, lichen

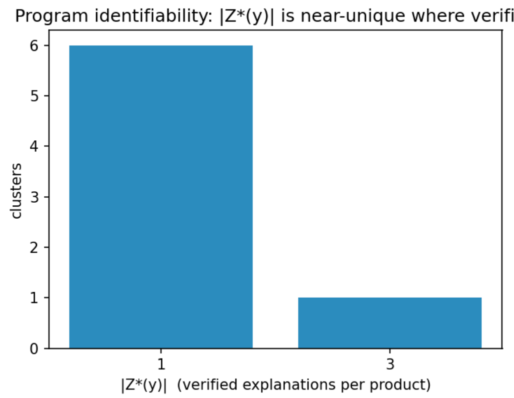

Figure 4: Explanation multiplicity for the seven tailoring-latent-verified clusters: six are uniquely explained and one has three consistent explanations. Where the joint search succeeds it is near-unique—the regime in which a verifier-constrained objective is well posed.

depsides. The exact membership is metric-dependent ($\sim$17 in the documented analysis), but the three classes are robust and are themselves a clean empirical statement about what "BGC$\to$structure" can mean for fungi.

Where coverage should extend. A structural classification of the 318 non-exact-reconstructible products locates the coverage frontier empirically. The unreachable set is dominated by small monocyclic aromatics ($\sim$100) and small aliphatics/lactones ($\sim$58), together with N-containing PKS–NRPS hybrids and non-PKS skeletons ($\sim$98); the polycyclic anthrone/anthraquinone/xanthone class often invoked as the cyclization frontier is only $\sim$ 16 clusters, mostly large ($C_{18–30}$) and largely coincident with the in-principle rearrangement ceiling above (sterigmatocystin, the cercosporin/elsinochrome perylenequinones). Extending the small-aromatic cyclization repertoire and modelling the PKS portion of hybrids would thus unlock substantially more than the higher PT-aromatic folds.

7 The conceptual result: grammar mismatch is the data deficit

The three tiers cohere into one statement. In colinear bacterial PKS the per-module core is a function of domain content, so it can be tabulated—ClusterCAD is the existence proof. In iterative fungal PKS the *same* domains must do different chemistry on different cycles, so the per-step core is not a function of domain content and no fungal ClusterCAD can be built; the end-product$\to$program map is many-to-one and the per-cycle intermediates are unobserved. This is why augmenting scarce fungal data with abundant bacterial data does not help: the bacterial data teaches a control flow that is structurally wrong for fungi. The exact-match 8/326, the 304/326 unknown subclasses, and the 95/99 core-unreachable diagnosis are three faces of the same fact.

Two orthogonal walls, not one. The ceiling is not uniform, and conflating its parts overstates the computability claim. A structural subclass proxy[1] (aromatic-carbon fraction and saturation)

[1]Per product: $N > 0$ heavy atoms $\Rightarrow$ hybrid/non-PKS; else aromatic-carbon fraction $f_{\mathrm{ar}} \geq 0.4 \Rightarrow$ NR/PR aromatic; $f_{\mathrm{ar}} \leq 0.1$ with H:C $\geq 1.4 \Rightarrow$ HR-reduced; otherwise PR/mixed. On the twelve curated systems this recovers the reduced-vs-aromatic axis for all four highly-reducing entries; it does *not* separate NR from PR (both aromatic), which

splits the 326 products into three regimes. Non- and partially-reducing aromatics ($\sim$61%) have an architecturally fixed reduction program—a non-reducing synthase makes an all-keto chain—so their only hard step is cyclization regiochemistry, which is deterministic and *incrementally addressable by executor coverage*: generalising the resorcylic-aldol closure to an arbitrary starter alkyl, for instance, lifts exact reconstruction from 8 to 9/326 (olivetolic acid) with no loss of soundness. Highly-reducing polyketides ($\sim$9%) cyclize trivially but carry the genuine computability wall—the per-cycle reduction program, i.e. the 100% $\to$ 57% gap, shown above to be neither positional nor bacteria-transferable. The remaining $\sim 30\%$ are N-containing PKS–NRPS hybrids or non-PKS skeletons, a scope boundary rather than a wall: the achiral C/H/O executor tracks the polyketide backbone up to the PKS–NRPS handoff (the linear checkpoint) but does not model the amino-acid condensation and release, so the PKS-core portion of a hybrid such as the tenellin synthase TenS stays computable even when its tetramic-acid product does not. Hence "fungal BGC→structure is uncomputable" is too strong: the computability wall is the narrow highly-reducing slice, while the dominant aromatic fraction is a tractable coverage problem. Because the executor is a *sound generator*, a cluster's catalytic alphabet moreover bounds a small enumerable set of producible cores—tiny when reduction is architecturally fixed, combinatorial for highly-reducing programs (a $\sim 13\times$ candidate-set ratio in our curated test)—which can be ranked and matched against metabolomics, recasting the impossibility result as a constrained hypothesis generator for genome mining. Conditioning that generator on the two observables a metabologenomics pipeline actually supplies—the domain alphabet and the MS-measured molecular formula (which pins the reduction *count*)—collapses our curated reachable set to a median of *one* candidate core (top-3 recall 9/10); the per-cycle reduction combinatorics, and with them the 0.57 grammar wall, largely dissolve when the domain alphabet and the mass are combined (the mass alone leaves larger cores with formula-isomers to rank, so both observables matter). The wall is thus partly an artefact of blind BGC→structure prediction: for small cores within executor reach, alphabet+mass is near-deterministic; and the larger cores that retain formula-isomers are *fragmentation*-distinguishable (a crude single-bond-cleavage fingerprint already gives all 112 formula-isomers of one C_{12} core distinct fragment sets), so predicted MS/MS resolves the residual—a domain→mass→fragmentation pipeline grounded throughout in the sound verifier. The binding constraint is then cyclization *coverage*, not computability.

8 A path forward: differential learning, demonstrated

If absolute prediction (BGC→structure) is fungal-data-starved, the *differential* question—given two modules and the difference in their domains, predict the difference in their intermediate structures—is abundantly supervised, because ClusterCAD provides thousands of bacterial modules with per-module intermediate structures. The per-step chemistry a domain performs transfers across bacteria and fungi (a ketoreduction is a ketoreduction); only the iteration grammar differs. A differential model learns the transferable half directly.

Setup. We scrape ClusterCAD (183 clusters → 1,752 modules → 1,091 clean extension modules), read per-step labels *from structure* (reduction state and KR stereochemistry at the active thioester), and train a small frozen model with domain-content features and two heads (reduction state; KR stereo), leave-cluster-out, against the antiSMASH-style domain rule and a majority baseline.

Domains under-determine per-step chemistry—even in bacteria. Structure-derived reduction state agrees with the domain rule only 80.8% of the time over the 1,091 modules; the

the two-walls split does not require.

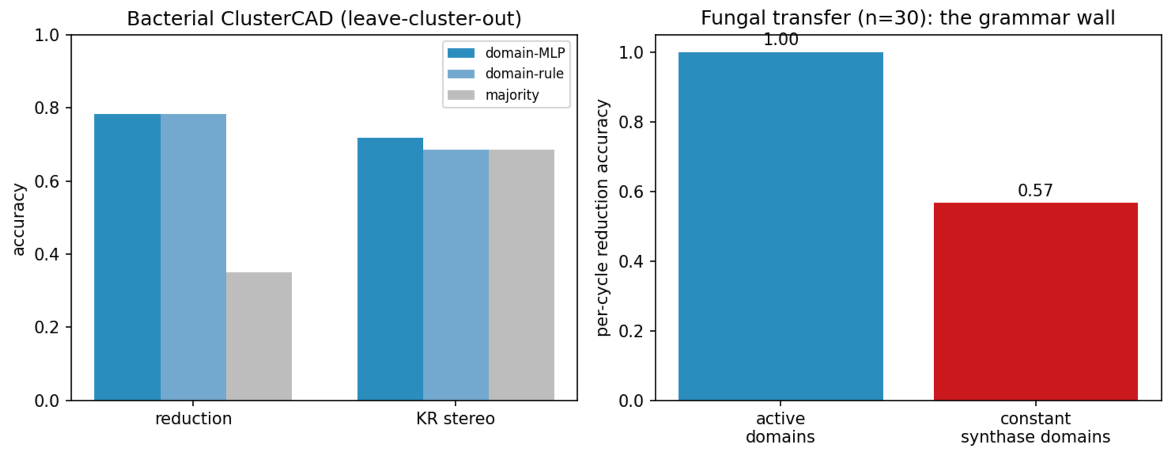

Figure 5: Differential PoC. Left: bacterial ClusterCAD (leave-cluster-out)—per-step reduction chemistry is learnable (domain-MLP $\approx$ domain-rule $\gg$ majority), while KR stereo is not domain-predictable. Right: fungal transfer—per-step chemistry transfers perfectly with active domains (1.00); given only the synthase's constant domain set it collapses to 0.57, the iteration-grammar wall.

$\sim$19% gap is real domain-present-but-inactive events (a DH or KR that does not fire). This is the LovB-degeneracy phenomenon quantified at scale, in colinear bacteria.

Results (Figure 5). The reduction head reaches 0.78, equal to the domain rule and far above majority (0.35); DH is perfect, while KETO recall is only 0.30 (the inactive-domain ceiling). KR stereochemistry is *not* domain-predictable (0.72 $\approx$ 0.69 majority): it requires sequence motifs. The headline is the fungal transfer: applying the bacteria-trained reduction head to 30 curated fungal cycles, accuracy is 1.00 **when the active domains are known** (per-step chemistry transfers perfectly) but 0.57 **given only the synthase's fixed domain set**. *That* 100%$\to$57% *gap is the iteration grammar*: the same genes must do different chemistry on different cycles, and gene content cannot say which.

The residual gap is not positional. One might hope the missing per-cycle signal is the running intermediate's state, cheaply proxied by chain *position* (cycle index, distance from the terminus)—available at mining time and, unlike the active-domain oracle, not a leak of which domains fired. It is not. Conditioning the bacteria-trained head on position does not beat the constant-domain wall on the 30 fungal cycles (leave-one-synthase-out 0.53 versus the wall's 0.57; bacteria-trained zero-shot, 0.57). The cause is mechanistic: bacterial per-step reduction is *positionally flat*—a function of domain content, not of chain position, across all 1,091 modules—so there is no positional gate for a bacteria-trained model (positional or geometric) to transfer. Within-fungal positional structure does exist but is subclass-specific and does not generalise across synthases (partially-reducing cycles benefit; highly-reducing cycles are hurt). The residual gap therefore demands *sequence or structure* signal, not geometry imported from bacteria. This is a small-sample probe (n = 30 cycles, 9 synthases), but the bacterial-flatness result it rests on spans all 1,091 modules.

A planned head, blocked on data access. The natural next test—does *sequence* carry the chemistry signal that gene content lacks?—is implemented as a frozen small ESM-2 (Lin et al., 2023) embedding with a tiny head, with two bacterial targets (KR stereo; inactive-domain detection) and the labelled set assembled (1,039 KR-bearing modules, 77 present-but-inactive KRs, 331 stereocentres). We report it honestly as *built but blocked*: ClusterCAD's per-domain sequence endpoint returned valid data on a first call and then reliably failed (HTTP 404) for bulk retrieval. We further attempted to reconstruct the per-domain sequences from primary data—MIBiG locus accessions → NCBI protein translations → a Pfam ketoreductase profile (PF08659) via HMMER—but the label→sequence join does not come clean: the structure-derived ClusterCAD labels count only confirmed C_2/C_3 extension modules, whereas the profile detects *every* ketoreductase domain (loading modules, standalone tailoring short-chain dehydrogenases, contig neighbours), reconciling for only a handful of clusters without reimplementing ClusterCAD's module parser. This is a deliberate algorithmic boundary rather than a missing experiment: the per-domain label→sequence link *is* the bespoke module decomposition ClusterCAD performs, and rebuilding it from raw genome sequence is out of scope here. The head is specified, coded, and runnable the moment a per-domain-labelled sequence source exists (a ClusterCAD export, or domain calls reconciled to ClusterCAD's module boundaries).

9 Related work

Riedling et al. (2024) establishes the 51–68% ceiling and the near-flat cross-domain result that motivates this work. Rule-based detectors and product predictors—antiSMASH/fungiSMASH (Blin et al., 2023), PRISM (Skinnider et al., 2020), DeepBGC (Hannigan et al., 2019), GECCO (Carroll et al., 2021), CLOCI (CLOCI, 2024)—detect clusters and, for bacteria, read off colinear products; none induce an iterative program. ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) is the modular-PKS intermediate database we both consume (for the differential PoC) and cite as the bacterial existence proof of contribution 4. MIBiG (MIBiG, 2023) is our pair source; BGC family methods such as BiG-SLiCE / BiG-FAM (Kautsar et al., 2021) define the within-family neighbourhoods relevant to differential pairs. Property/arrangement predictors such as CHAMOIS (CHAMOIS, 2025) infer ontology-level chemistry, not exact iterative programs. Methodologically we draw on execution-guided, neurosymbolic program synthesis (Ellis et al., 2021) and on protein language models (Lin et al., 2023); chemistry is handled with RDKit (Landrum, 2023). Our distinguishing stance is to *measure* exact reconstructibility and to ground every claim in a sound deterministic verifier.

10 Limitations

- **Partial executor coverage.** Single-mode cyclization plus a tetraketide-aromatic and a PT-naphthalene class (skeleton-correct; hydroxyl regiochemistry flagged) are implemented; the remaining PT-aromatic classes, the Diels–Alder, and macrolactonization are not. Thus the 95/99 core-unreachable count is partly executor coverage and partly true rearrangement.
- **Additive-only tailoring.** The $\sim$12 edit types are additive and the joint verifier is "exact-by-budget" (skeleton embedding + Δ-formula + gene/parsimony budget), threshold-free but not yet full atom-by-atom positional reconstruction (positions affect $|Z^\star(y)|$ multiplicity, not existence).
- **Gene budget over-counts** when a locus accession is a whole contig (non-cluster genes included); HMMER on cluster-scoped sequences would tighten it.
- **Subclass unknown for** 304/326; stereochemistry is not modelled (the executor is achiral by design); numbers are over MIBiG-deposited entries, biased toward well-studied (often heavily

tailored) compounds.

- **No trained genome→structure model is claimed.** The differential reformulation demonstrates the transferable per-step chemistry; the iteration grammar (the 100% → 57% gap) remains open for *blind* prediction, though we show it is largely circumvented given an MS mass (Section 7).
- **The mining pipeline is demonstrated, not yet validated.** The alphabet+mass+MS/MS chain is shown as near-unique recovery on reachable cores and as fragment-fingerprint *separability* under a crude in-silico fragmenter; validation against real untargeted-metabolomics spectra (e.g. CFM-ID / GNPS), per-cluster domain alphabets from antiSMASH/fungiSMASH, and a learned candidate-ranking prior (data-limited at present) are the natural next steps.

11 Conclusion

The fungal BGC-to-molecule ceiling is, we have argued and measured, a grammar problem masquerading as a data problem—and grammar mismatch is itself the data deficit, because iteration makes the per-step core uncomputable from gene content. We quantified the reconstructible fraction (exactly 8/326; 7 of the 106 gene-budgeted clusters under a tailoring-latent model, but 64% conditional on a producible core), showed that structural-similarity proxies are unreliable for this question, and located the wall precisely at PKS-core computability. The path forward is differential: per-step chemistry is learnable from abundant bacterial data and transfers to fungi perfectly when the active domains are known—and the residual 100% → 57% gap is a clean, quantitative target for future work on inducing the iteration program itself—and a *narrow* target: the ceiling decomposes into a small highly-reducing computability wall and a broad, coverage-addressable aromatic-cyclization wall, and the same sound executor serves as a constrained candidate generator, turning an impossibility result into a usable mining tool. The tools, benchmark, and analyses are open and reproducible.

Reproducibility. All results are produced by single commands over pinned dependencies; see Appendix A.

A Reproducibility

Python 3.12, RDKit 2025.9.6; optional `torch`/`transformers` for the differential and ESM-2 components. Each result maps to one command: `make data` (the 326 pairs), `make roundtrip` (executor gate), `make search` (the verifier and $Z^\star(y)$), `make reachability` (exact-match 8/326), `make corematch` (the core-match sensitivity), `make phase0c` (tailoring-latent 7/106), `make clustercad` and `make differential` (the differential PoC), `make figures` (Figures 2–5), and `make test` (58 tests). Per-result numbers, scripts, and git commits are tabulated in `PAPER_READINESS.md`.

References

O. Riedling, A. K. Walker, et al. Predicting fungal secondary metabolite activity from biosynthetic gene cluster data using machine learning. *Microbiology Spectrum*, 2024.

K. Blin et al. antiSMASH 7.0. *Nucleic Acids Research*, 2023.

M. A. Skinnider et al. Comprehensive prediction of secondary metabolite structure and biological activity (PRISM 4). *Nature Communications*, 2020.

C. H. Eng, T. W. H. Backman, et al. ClusterCAD: a computational platform for type I modular polyketide synthase design. *Nucleic Acids Research*, 46(D1):D509–D515, 2018.

ClusterCAD 2.0: an updated computational platform for chimeric type I PKS and NRPS design. *Nucleic Acids Research*, 2023.

G. D. Hannigan et al. A deep learning genome-mining strategy for biosynthetic gene cluster prediction (DeepBGC). *Nucleic Acids Research*, 47(18):e110, 2019.

L. M. Carroll et al. Accurate de novo identification of biosynthetic gene clusters with GECCO. *bioRxiv*, 2021.

CLOCI: unveiling cryptic fungal gene clusters with generalized detection. *Nucleic Acids Research*, 52(16):e75, 2024.

The MIBiG consortium. MIBiG: minimum information about a biosynthetic gene cluster (release 4.0). *Nucleic Acids Research*, 2023.

S. A. Kautsar et al. BiG-SLiCE / BiG-FAM: clustering biosynthetic gene clusters at scale. *GigaScience / Nucleic Acids Research*, 2021.

CHAMOIS: inferring metabolite properties from protein domains in biosynthetic gene clusters. 2025.

S. M. Ma, J. W.-H. Li, J. C. Vederas, Y. Tang, et al. Complete reconstitution of a highly reducing iterative PKS (lovastatin nonaketide synthase LovB/LovC). *Science*, 326(5952):589–592, 2009.

J. M. Crawford et al. Structural basis for biosynthetic programming of fungal aromatic polyketide cyclization (the product-template domain). *Nature*, 461:1139–1143, 2009.

K. Ellis et al. DreamCoder: bootstrapping inductive program synthesis with wake-sleep library learning. *PLDI*, 2021.

Z. Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model (ESM-2/ESMFold). *Science*, 379(6637):1123–1130, 2023.

G. Landrum et al. RDKit: open-source cheminformatics. `https://www.rdkit.org`.